EEE

1. write the 8 bit EBCDIC code for the following characters, A, F ( ) =
2. perform subtraction using 2's complement method $(263)_{10} - (2F)_{16}$
3. Convert $(512.0625)_{10}$ to base 4, base 8 and base 16
4. write the binary code for the gray code 10110111.
5. state and prove De Morgan's theorems with the help ~~for~~ of truth tables.
6. show that both NAND and NOR gate can be use as an universal gate.
7. Simplify the following Boolean expression.
   (i) $\overline{A\bar{B}} + \bar{C}D + EF$
   (ii) $\bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$
8. Implement a full adder using two half adder
9. Minimize the following function using Quine-Mc cluskey method (up to level one)
   $f(a,b,c,d) = \Sigma_m (0,1,2,4,6,8,12,14)$

10. what do you mean by (i) Fan out and (ii) Noise margin

11. what are difference between transistor and FET.

12. Draw and explain the circuit operation of 2 digit TTL NOR gate.

13. Explain the characteristics of CMOS.

14. Discuss the operation of a 555 timer based bistable multivibrator.

15. Draw the diagram of 555 timer and explain

16. Describe the operation with diagram of R-2R Ladder D/A converters.

17. what are specification of ADCs.

18. Draw the block diagram of Dual slope A/D converter and discuss it's operation.

19. Write the ~~operation~~ application of ADC and D/A converter.

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20. Distinguish between latch and flip-flop

21. Explain clocked SR flip-flop with application.

22. Draw the circuit diagram of D Latch and explain it's operation.

23. Explain the method of edge triggering.

24. Explain master slave JK flip Flop with timing diagram indetail.

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